What Matters in Relational-to-Ontology Learning? Data Evidence, Completeness Guard, and Scoring — Extended Results

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This report collects the details behind the paper *What Matters in Relational-to-Ontology Learning? Data Evidence, Completeness Guard, and Scoring*: the study design and cohorts, additional results of the main study, the paired experiment, the later round, the audit of the converted gold mappings, baselines, configurations and resources. Sections, tables and figures numbered with the prefix S are in this report; all other numbers refer to the paper.

S1. Study Design, Cohorts and Denominators

Fig. S1 summarises the study design of the paper: which data play which role, which systems we compare, how we measure them and what each research question found.

Table S1 lists every cohort behind a number in this paper. Counts from different rows must not be combined: for example, the 58 self-consistency cases, the eight held-out databases and the 12 tables of the probe are different units.

Separate tracks: Later work extended the study to 11 databases and 13 whole-schema configurations. Its renaming, key-removal, sample-row, completeness and model-grid results are reported in Section S4, with the eight databases of the main study as the primary scope. A separate per-table run of three further vendor configurations reached 94 of 99 planned database samples, with no imputation of missing outputs; the two complete configurations are reported descriptively in Section S10, and the third (Fable, 28 of 33) is not averaged. Three of the added databases raise gold questions that do not concern the eight-database results, although they enter the eleven-database means of Sections S4 and S10. An OMOP conversion maps 58 of its 63 data properties to the primary-key column of their subject, which changes the meaning of attributes such as gender, and the MusicBrainz gold identifies entities by the public identifier `gid`, whereas its tables are keyed by the internal `id`. We changed no denominator of this paper because of them. A licence inventory of 12 databases underlies the data-availability statement of Section VII. A screening for new large databases admitted none. PNM v6 was converted only as a diagnostic: of its 34 physical tables, 30 are referenced by its mapping and 3 of those are missing from the release, and all 327 of its mapping assertions are inexpressible in Burr's format, which leaves no gold elements. Two further candidates, WHOW and MolMeDB, were declined for this round. Foreign-key target-ranking results on other schemas belong to a different task, with different inputs, labels and units; they enter none of our tables, candidate pools or dataset counts. Finally, a pilot that fine-tuned a local 4B model on 38 development examples from three scenario families showed signs of overfitting (validation loss rose from 0.120 after 50 steps to 0.322 after 200) and was scored with a reference-exact metric that differs from Burr F1; it is not a Burr result.

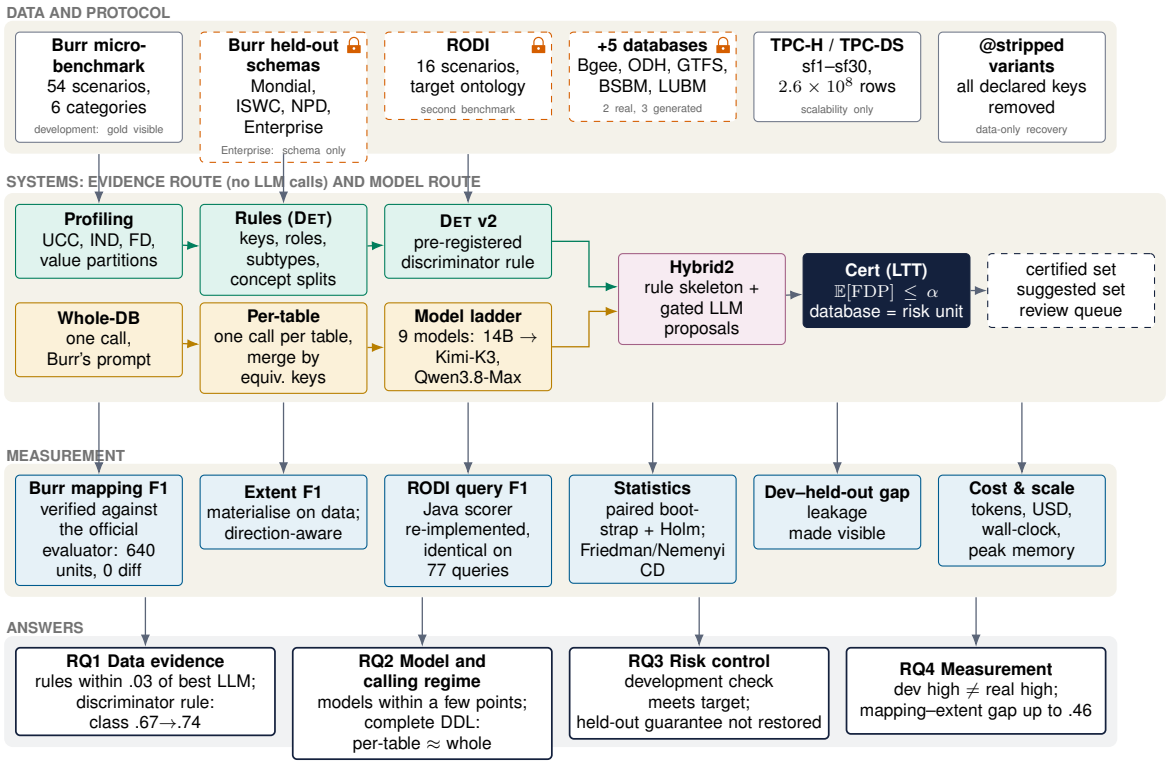


Fig. S1. Study design. Gold mappings of the held-out schemas (dashed red) are sealed during system design. Two routes, evidence rules and LLM generation in two calling regimes, are scored with three metrics, paired statistics and the gap between development and held-out scores. The answer boxes summarise the results; "640 units, 0 diff" is the official-evaluator check of the mapping metric, and "no LLM calls" refers to model charges only, since profiling has a computational cost.

TABLE S1
COHORTS AND DENOMINATORS USED IN THIS PAPER, AND TRACKS KEPT SEPARATE.

Cohort	Composition and use
Development set	54 Burr micro-benchmark scenarios enumerated by our loader, six categories; rule and prompt design, CERT calibration
Burr held-out set	Mondial, ISWC, NPD (with data) and Enterprise (7 tables, schema-only); Table I
Eight held-out databases	Burr's three with data, plus Bgee and ODH (real data) and GTFS, BSBM and LUBM (generated); Table S4, Fig. 1
Self-consistency	58 cases (54 + 4) for the mapping metric; 57 for the extent metric, since Enterprise has no data
Official-evaluator check	640 scenario–system units over six arms (Section V)
Complete-DDL control	12 outputs: 2 models $\times$ 2 regimes $\times$ 3 samples (Table S5)
Paired complete-DDL experiment	Qwen and DeepSeek $\times$ 3 runs: Burr's three databases with data (198 prompts, 1,188 requests, 42 whole-database mappings) and the five added databases (53 prompts, 318 requests, 60 mappings); Kimi-K3 added later (181 complete prompts, 543 requests, 48 mappings); schema-only prompts (Section VI-B3)
Key-column control	8 whole-schema prompts with one added instruction $\times$ 3 models $\times$ 3 runs (72 requests and mappings), post hoc (Section S3)
CERT pools	46 sources (main, Fig. S12); variants with 49 and 27 sources
CREATE probe	12 of the 66 affected NPD tables $\times$ 3 conditions $\times$ 3 repeats; 108 completed of 216 planned, `qwen3.8-max` (Section S8)
Frozen replay	4,735 scoring and 9,673 coverage tasks; 11,886 files (Section VII)
Gold checks	All 332 retained entries of the five added databases executed against their source mappings; 233 of them audited by two model passes (Section S5)
Completion re-analysis	Archived outputs of the paired experiment re-merged under three guards, no new calls (Section S3)
Reasoning check	48 single calls of Qwen and Kimi-K3 with reasoning on, plus 3 diagnostic re-sends (Section S3)
RODI	16 scenarios (Section S2)
TPC-H, TPC-DS	sf1–sf30; profiling cost only, no gold mappings
Later round	Renaming (paraphrased and opaque names), keys removed and 0/5/20 sample rows on 8 databases (11 secondary); 13 whole-schema configurations including four Claude and three GPT models; census of 9,673 whole-schema outputs; $K = 3$ (Section S4)
Exploratory vendor track	Two configurations, 11 databases $\times$ $K = 3$ = 33 database samples each; means over the eight databases of Table S4 and over 11 databases; descriptive only (Section S10)
Not in this paper	Other results on the three further databases; licence inventory of 12 databases; PNM v6 conversion diagnostic (no gold elements); a foreign-key target-ranking subtask; a small-model pilot

TABLE S2
PRE-REGISTERED RULE CHANGES (DET-V2): DEVELOPMENT-SET EFFECT, HELD-OUT EFFECT AND DECISION. HELD-OUT MEANS ARE OVER THE THREE BURR DATABASES WITH DATA.

Change	Dev	Held-out	Kept
Discriminator-only subclasses	never fires	class .67→.74	yes
Name cue for self-ref./split	never fires	included above	yes
Orient parent→child	rel. .69→.75 (extent)	rel. .26→.19 (extent)	no
Early-stop uniqueness check	unchanged	unchanged, 1.9× slower	no

S2. ADDITIONAL RESULTS OF THE MAIN STUDY

This section collects results of the main study that support the paper but are not needed to follow it.

Further details: In the model ladder (Fig. 3), only one of 24 development-set differences from the DeepSeek reference is significant after Holm correction, the class F1 of the 14B model. In rule ablations, the link-table, subtype and two-concepts rules matter most: disabling them lowers relation F1 by .135 and class F1 by .059 and .052. On the development set, where no scenario has more than six tables, each model's per-table relations are 0.02–0.07 below its single call. A query-level check on five relation pairs from cached DET outputs agrees with the extent metric of Section VI-E: the three pairs whose predicate and direction match the gold return the same answers (2,285 of 2,285, 10 of 10 and 21 of 21 key pairs), while the two ISWC pairs with a different predicate each return one extra answer; these cases illustrate the effect, and how often it occurs is not known.

Development set and held-out databases: Fig. S2 breaks the development-set numbers of Table I down by element type, and Fig. S3 shows every held-out database. On Burr's held-out databases the two regimes are close: on the four-database mean, Kimi-K3 scores 0.62/0.54/0.57 in a single call and 0.60/0.52/0.55 per table, DeepSeek per table 0.55/0.49/0.51 and the 35B model per table 0.51/0.40/0.55. The one large difference is DeepSeek whole-schema generation (0.49/0.44/0.40), which comes from NPD: two of its five samples returned only 14 and 16 of the 70 tables, and a third stopped at 60. On Mondial and ISWC the regimes are within 0.12 of each other. Table S3 shows that sampling variance is an order of magnitude smaller than these differences; repeated calls with a fixed prompt are not independent replications, so the intervals say nothing about variation over databases.

Gain of decomposition by schema size: Fig. S4 shows the per-table minus whole-schema difference of the main study against table count (Section VI-B2).

Development versus held-out gap: Fig. S5 plots development against held-out scores (Section VI-D).

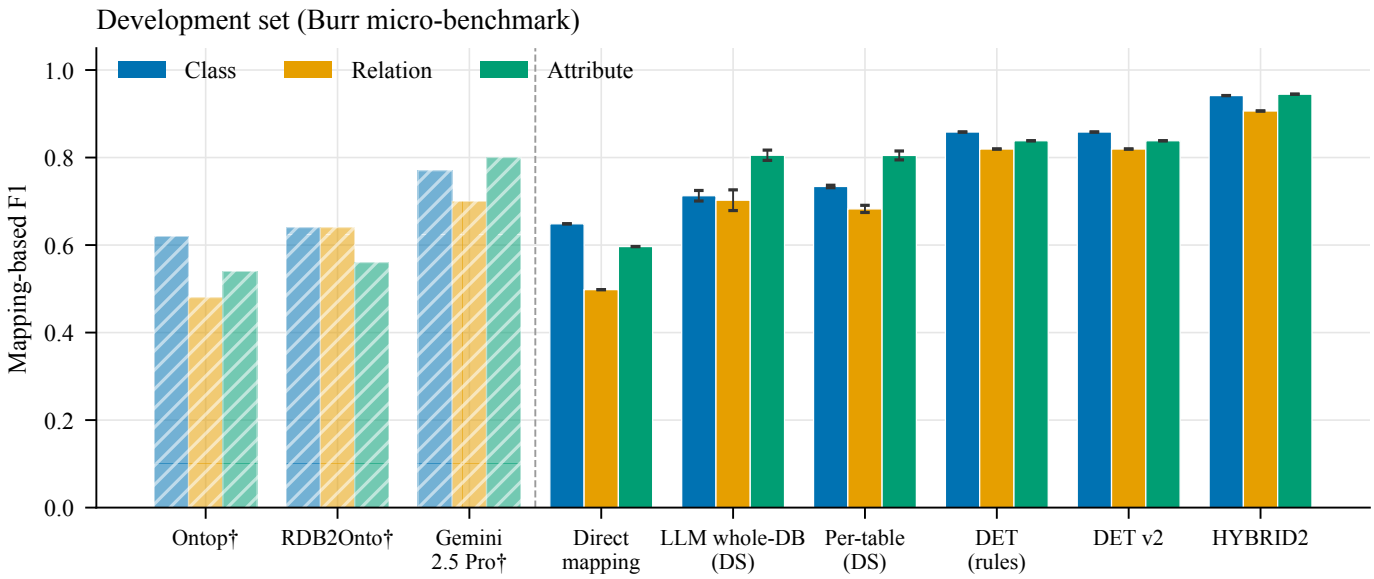
Scale stress test: After our initial analysis we pre-registered a stress test (Fig. S6): sub-databases of 10, 20 and 40 tables, grown along foreign keys from NPD and Mondial-FK with three seeds each, with the gold cropped to each sub-database. With the corrected prompt, DeepSeek reaches a three-way mean of 0.72/0.74/0.73 on 10/20/40 NPD tables and Kimi-K3 0.72/0.74/0.71. One DeepSeek run on 40 tables wrote a complete mapping in a format the scorer cannot parse; it scores 0, which lowers that mean to 0.65. Both models stay at 0.58–0.70 on every Mondial-FK sub-database. The first run of the stress test cut its sub-database prompts from the same faulty extraction, and Kimi-K3 dipped to 0.55 at 20 tables, on the one sub-database that contains the four surviving CREATE statements; with the corrected prompt the dip disappears. Under the pre-registered rule, neither model has a collapse point at any size. The stress test also exposed how per-table generation fails: under DeepSeek, two NPD tables with 65 and 79 columns always exceed the 8,192-token output cap per call, and 51 of 1,260 single-table calls returned nothing, which puts DeepSeek per-table 0.07–0.14 *below* whole-schema generation on the NPD sub-databases. Kimi-K3 writes shorter outputs and is level with whole-schema generation.

Complete-DDL control on the full NPD: Table S5 is a separate frozen-pipeline cohort of 12 outputs (two models, two regimes, three samples). Whole-schema generation recovers from 0.06/0.07 with the defective prompt to 0.70 for Kimi-K3 (three complete samples) and 0.59 for DeepSeek (two samples at 0.73, one that stopped after 20 tables). Re-running the full database with the corrected extractor in our production runs, which supply the whole-schema NPD values of Tables I and S4, gives the same picture: Kimi-K3 scores 0.71 in all three samples and DeepSeek 0.40, because two of its three samples stopped after 14 and 16 tables (0.50 over five samples). The control and the production runs are different sets of calls, so we report both and pool neither.

Ranking across all blocks: Fig. S8 shows the critical-difference diagram over the ten blocks. At the top are DET-V2 (mean rank 6.05), DET (6.80), the convention-tuned DeepSeek prompts (7.0–7.2, leakage), HYBRID2 (7.2) and the two Kimi-K3 configurations (8.2 and 9.1). At the bottom are DM, the simple prompt, RDB2Onto and Ontop (19.4–21.8). With ten blocks the critical difference is 12.0, so only the two ends can be told apart.

Second benchmark: RODI: RODI asks for a mapping to a *given* target ontology, so we evaluate two regimes (Fig. S9). In regime (a) our systems see only the database, and we align their output ontology to the target with RODI's own LogMap [1] step, as was done for BootOX. In regime (b) the LLM arms also receive the target vocabulary, as in LLM4VKG. In regime (a) all six systems lie within 0.35–0.40 query F1 over the 16 scenarios (PER-TABLE highest at .395); under LLM4VKG's grouping our best is .42, against the reported .48 for BootOX and .44 for IncMap. In regime (b), DeepSeek WHOLE-DB reaches .607 (.62 under the grouping), against .65 reported for GPT-4o. In regime (a), 61–68% of fully wrong queries come from target vocabulary that was never aligned and only 6–16% from mapping structure.

Scalability and cost: Profiling plus rules runs end to end on TPC-H [2] sf30 (2.6×10^8 rows) in about 14 minutes with a 26.8 GB peak (Fig. S10, which also shows TPC-DS [3]). These generated databases have no gold mappings, so they measure only the cost of profiling and key recovery. Full-table verification of unique column combinations takes 90%



† reported in Burr (SIGMOD 2026), Table 3.

Fig. S2. Development set (Burr micro-benchmark), mapping-based F1 by element type. Hatched bars are values reported by Burr. Error bars are standard deviations over LLM samples. Rules and hybrid lead here, but both were designed while looking at these gold mappings (Section III).

TABLE S3
PER-TABLE GENERATION, HELD-OUT FOUR-DATABASE MEAN OVER K SAMPLES: MEAN ± SD AND 95% t-INTERVAL. SAMPLING VARIANCE IS AN ORDER OF MAGNITUDE SMALLER THAN THE DIFFERENCES BETWEEN REGIMES AND BETWEEN DATABASES. THE INTERVALS DESCRIBE REPEATED CALLS WITH A FIXED PROMPT ON THE SAME FOUR DATABASES; SUCH CALLS ARE NOT INDEPENDENT REPLICATIONS, AND THE INTERVALS SAY NOTHING ABOUT VARIATION OVER DATABASES. NPD INPUTS AS IN SECTION VI-B1.

Model	K	Class	Relation	Attribute
DeepSeek Flash	10	.546±.015 [.535,.556]	.489±.009 [.483,.495]	.507±.025 [.489,.526]
Qwen3.6-35B-A3B	6	.505±.005 [.500,.511]	.400±.012 [.388,.412]	.552±.005 [.548,.557]
Kimi-K3	6	.604±.004 [.600,.608]	.519±.006 [.513,.525]	.549±.002 [.548,.551]
Qwen3.8-Max	6	.521±.014 [.506,.536]	.451±.023 [.428,.475]	.557±.006 [.550,.563]

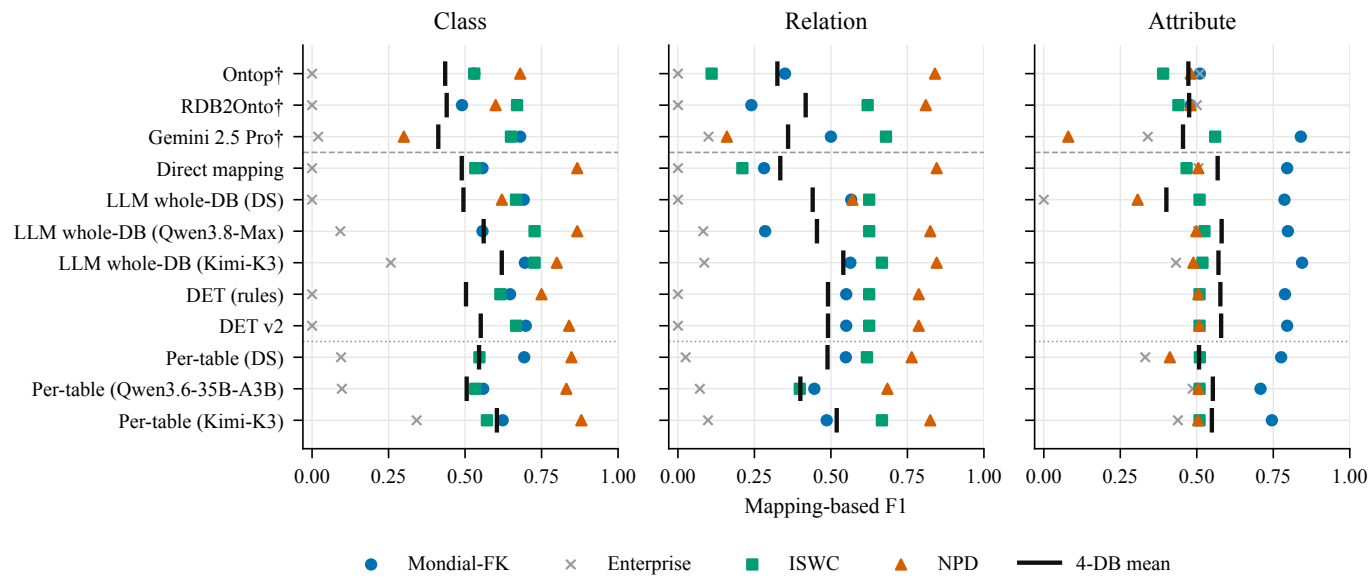
TABLE S4
EIGHT HELD-OUT DATABASES WITH DATA: MAPPING-BASED F1 (CLASS/RELATION/ATTRIBUTE), MEAN OVER DATABASES. LLM ARMS USE $K = 3$; VALUES ARE EQUALLY WEIGHTED MEANS OF PER-DATABASE MEANS. PER-TABLE AND HYBRID2 NPD VALUES COME FROM THE DEFECTIVE BUILDER.

System	8 DBs	Burr 3	Added 5
DM	.59/.43/.61	.65/.45/.59	.55/.42/.62
DET	.61/.56/.63	.67/.65/.60	.57/.51/.65
DET-V2	.63/.56/.63	.74/.65/.60	.57/.51/.65
HYBRID2	.57/.54/.62	.63/.65/.60	.53/.47/.64
WHOLE-DB DeepSeek	.59/.56/.62	.61/.54/.53	.57/.57/.68
WHOLE-DB 35B-A3B	.63/.56/.63	.72/.50/.55	.57/.59/.68
WHOLE-DB Kimi-K3	.64/.61/.66	.74/.69/.62	.57/.56/.69
PER-TABLE DeepSeek	.62/.58/.64	.70/.65/.56	.57/.54/.68
PER-TABLE 35B-A3B	.58/.53/.64	.64/.51/.57	.55/.54/.68
PER-TABLE Kimi-K3	.62/.59/.65	.69/.66/.59	.58/.54/.68

TABLE S5
FULL NPD, FROZEN PIPELINE, $K = 3$. THREE-WAY MEAN F1 OF WHOLE-SCHEMA GENERATION WITH THE DEFECTIVE PROMPT (4 OF 70 CREATE STATEMENTS) AND WITH THE COMPLETE DDL (CONTROL, EXPLORATORY); NUMBER OF CLASSES PREDICTED BY WHOLE-SCHEMA GENERATION IN EACH SAMPLE (80 IN THE GOLD); AND PER-TABLE GENERATION FOR REFERENCE. THE CONTROL IS AN INDEPENDENT COHORT OF 12 OUTPUTS (TWO MODELS × TWO REGIMES × THREE SAMPLES), SEPARATE FROM THE PRODUCTION RUNS BEHIND TABLES I AND S4; VALUES FROM THOSE RUNS ARE NOT SUBSTITUTED HERE. "COMPLETE" IDENTIFIES THE WHOLE-SCHEMA DDL CONDITION. THE PER-TABLE REFERENCES USE THE DEFECTIVE BUILDER, WHICH OMITS 66 OF THE 70 NPD CREATE STATEMENTS, SO THIS IS NOT A COMPLETE-DDL COMPARISON OF THE TWO REGIMES; TABLE II IS.

Model	DDL in prompt	WHOLE-DB	Classes	PER-TABLE
DeepSeek Flash	4 of 70 tables	.06	4	.68
DeepSeek Flash	complete	.59	70/70/20	.68
Kimi-K3	4 of 70 tables	.07	4–5	.74
Kimi-K3	complete	.70	70/70/70	.72

of the time, and key recovery without declared keys stays stable as scale grows (8/8 primary keys at sf1 and sf10, 7/8 at sf30). Fig. S11 plots cost against quality for the main study: the frontier has two points, DET-V2 without model calls (0.611) and single-call Kimi-K3 (0.637 at about 57k tokens per database sample), while per-table generation spends 370k–510k tokens for 0.585–0.619. According to the run records of the main study, its DeepSeek calls cost about $67, and seven further models received 7,317 calls with about 44M



† reported in Burr (SIGMOD 2026), Table 3.

Fig. S3. Held-out real databases. Each marker is one database, and the bar is the four-database mean. Whole-schema NPD values use the corrected prompt; per-table NPD values carry the input caveat of Section VI-B1. In these runs the two calling regimes are close on every database. DeepSeek whole-schema generation on NPD (triangle) is pulled down by two incomplete samples out of five. †: published values; for Mondial and NPD they predate the authors' correction (Section S6).

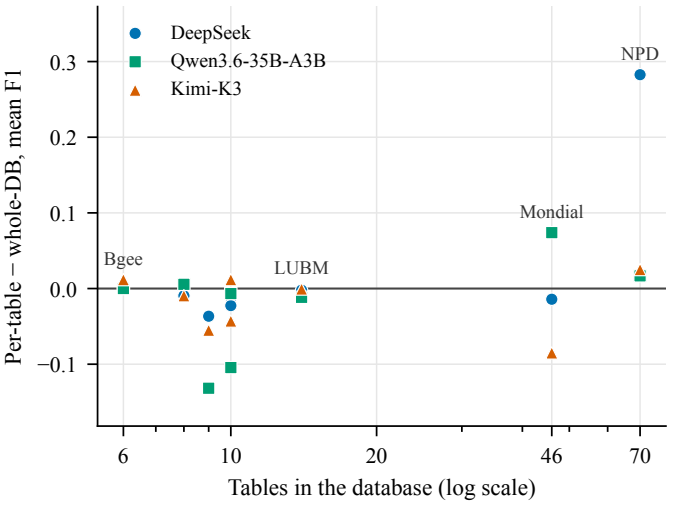


Fig. S4. Gain of per-table over whole-schema generation (same model, three-way mean F1) against the number of tables on the eight held-out databases (main study). The gain lies between -0.13 and $+0.08$ everywhere except DeepSeek on NPD, where two of five single-call samples were incomplete. Per-table NPD values come from the defective builder.

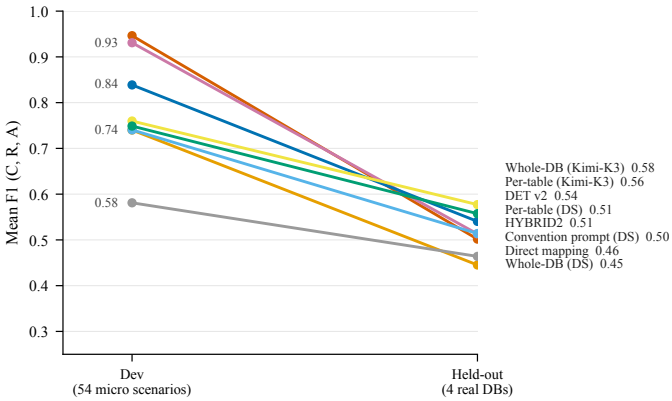


Fig. S5. Development versus held-out three-way mean F1. Convention-tuned prompts and the hybrid drop most (0.41–0.44). Single-call Kimi-K3 and per-table generation drop least (0.18–0.23).

input tokens; these are historical records and have not been reconciled with an invoice.

Risk-control curves: Fig. S12 shows the realised false-discovery proportion against the target α discussed in Section VI-D. More sources do not help: a pool of 49 sources from 15 model families still reaches a class FDP of .329 on the real databases.

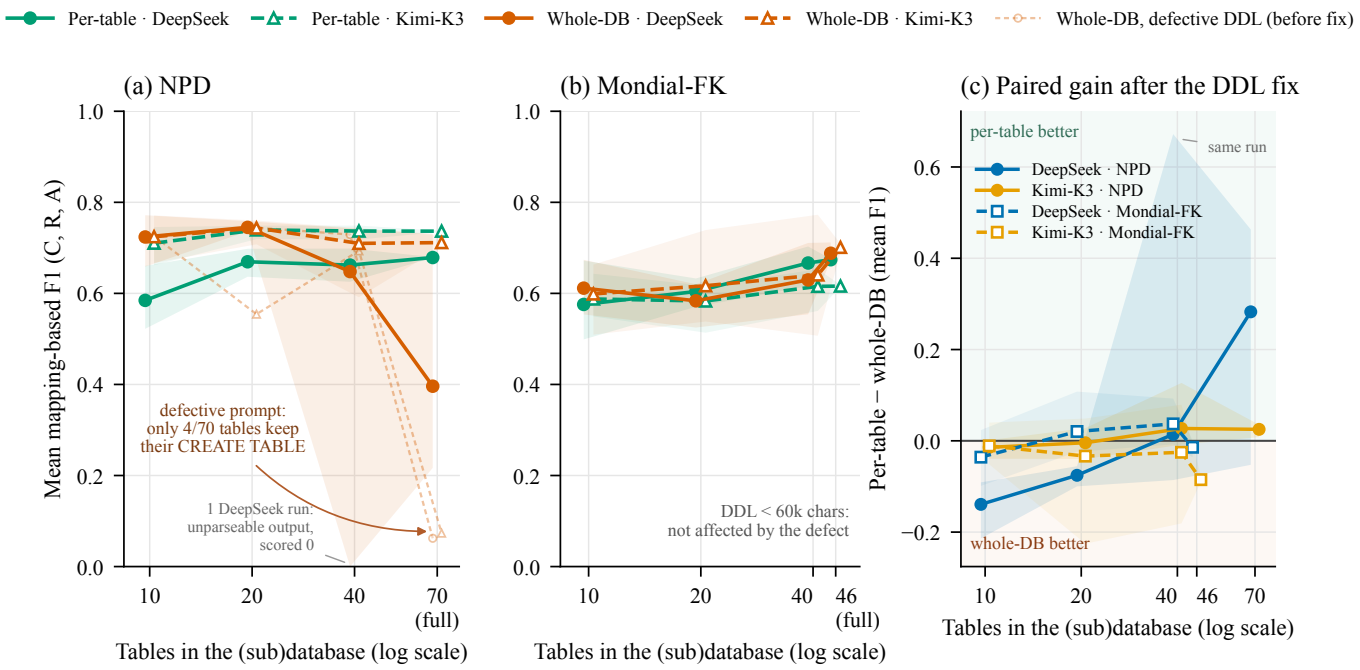


Fig. S6. Scale stress test (pre-registered). (a) NPD and (b) Mondial-FK sub-databases of 10, 20 and 40 tables (three seeds, $K = 3$) and the full database. Solid lines use the corrected prompt; bands span seeds and samples. Thin dashed lines show whole-schema generation under the faulty prompt, which kept the CREATE statements of only 4 of the 70 NPD tables. With the corrected prompt, Kimi-K3 holds 0.71–0.74 at every size. DeepSeek holds up to 40 tables and drops on the full NPD because two of its three runs stopped after 14 and 16 tables. One DeepSeek run on 40 tables is unparseable and scores 0. Mondial-FK is below the 60k-character threshold and unaffected. (c) Paired gain of per-table over whole-schema generation after the whole-schema fix. Per-table points use the defective builder (Section VI-B1); we have not audited its prompts for the sub-databases.

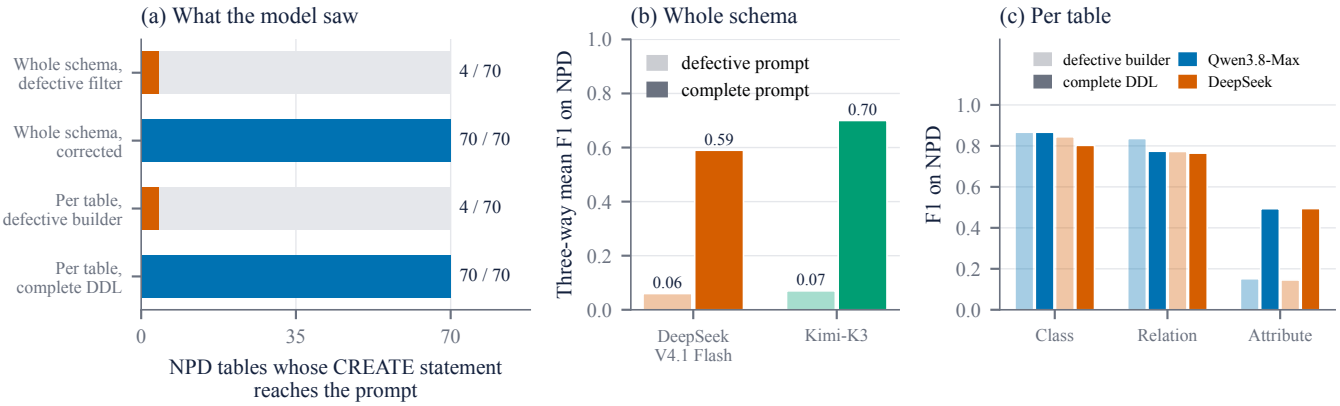


Fig. S7. What the model saw on NPD and what it cost (moved from the paper). (a) NPD tables whose CREATE statement reaches the prompt, with the defective and the corrected whole-schema filter and with the defective and the complete per-table builder. (b) Whole-schema generation: three-way mean F1 on NPD with the defective and the complete prompt (frozen pipeline, $K = 3$; Table S5). (c) Per-table generation in the paired experiment (Section VI-B3), whose prompts carry no sample rows: with the defective builder the model sees only key columns, and attribute F1 falls to 0.15; with complete DDL it is 0.49.

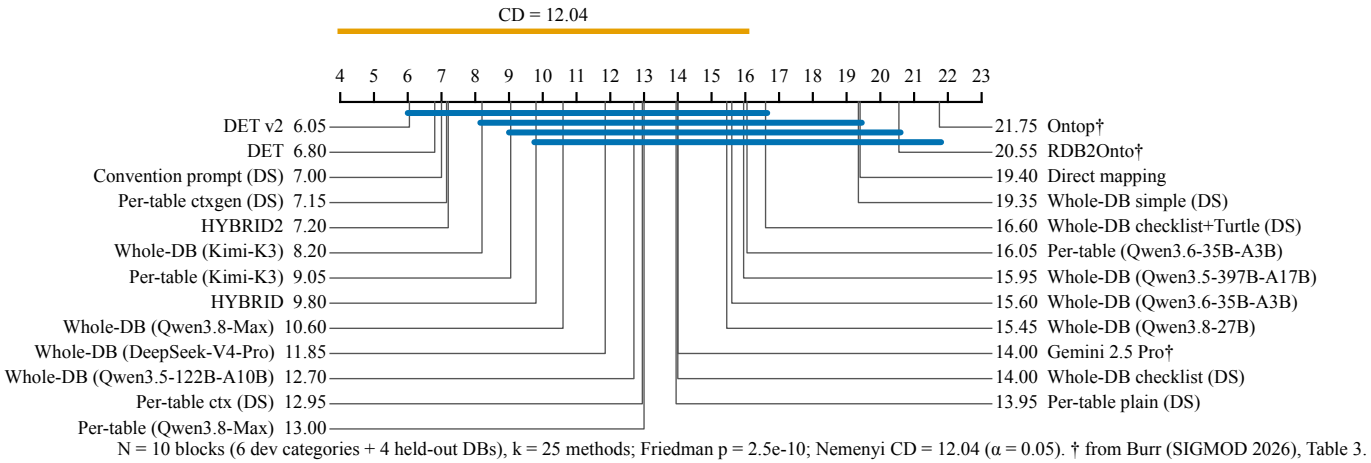


Fig. S8. Nemenyi critical-difference diagram over ten blocks (six development categories and four held-out databases), 25 systems. Lower rank is better. †: Burr-reported systems, ranked with their published values; the authors' later Mondial and NPD corrections (Section S6) are not included, and ranks were not recomputed. Per-table NPD blocks carry the input caveat of Section VI-B1.

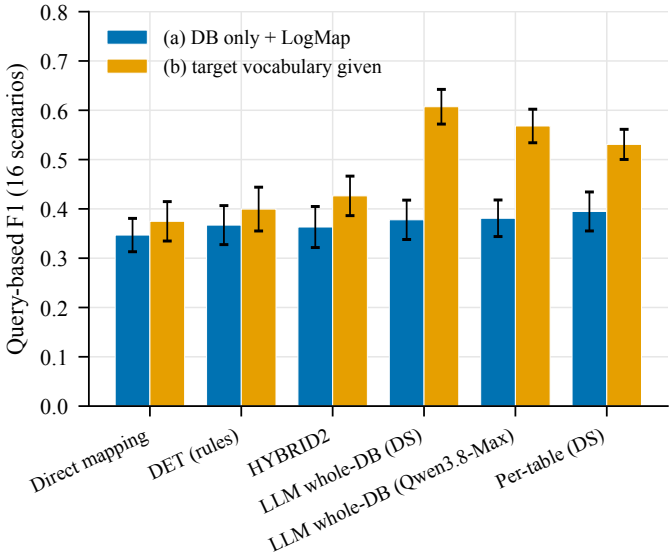


Fig. S9. RODI, query-based F1 over 16 scenarios (mean ± s.e.). Regime (a): database only, LogMap alignment. Regime (b): target vocabulary given.

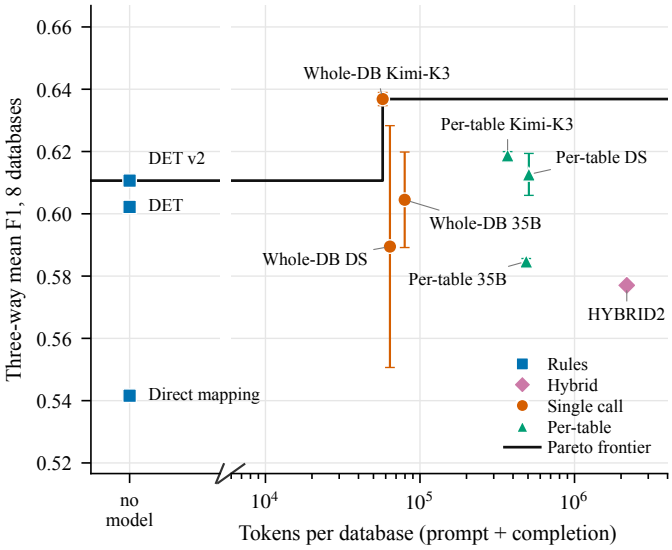


Fig. S11. Cost–quality trade-off of the main study (run records). x: prompt plus completion tokens per database sample, averaged over Burr's three databases with data (rules use no model). y: three-way mean F1 over the eight held-out databases, with bars over $K = 3$ samples. The two axes therefore cover different database sets, and tokens are not prices. Within this cohort the Pareto frontier runs from DET-V2 (no model call) to single-call Kimi-K3. Per-table generation costs 6–8× the tokens of a single call with no measurable gain, and HYBRID2 about 30×. Later exploratory runs (Sections S10 and S11) are not included, and no frontier is claimed for them.

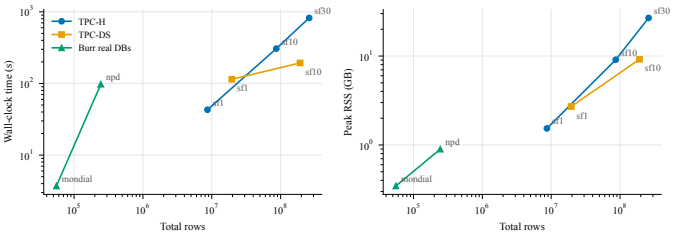


Fig. S10. Scalability of profiling plus rules (`@stripped`, default budgets): wall-clock time and peak memory against total rows, log–log. TPC-H and TPC-DS are generated databases without gold mappings; this figure reports cost only.

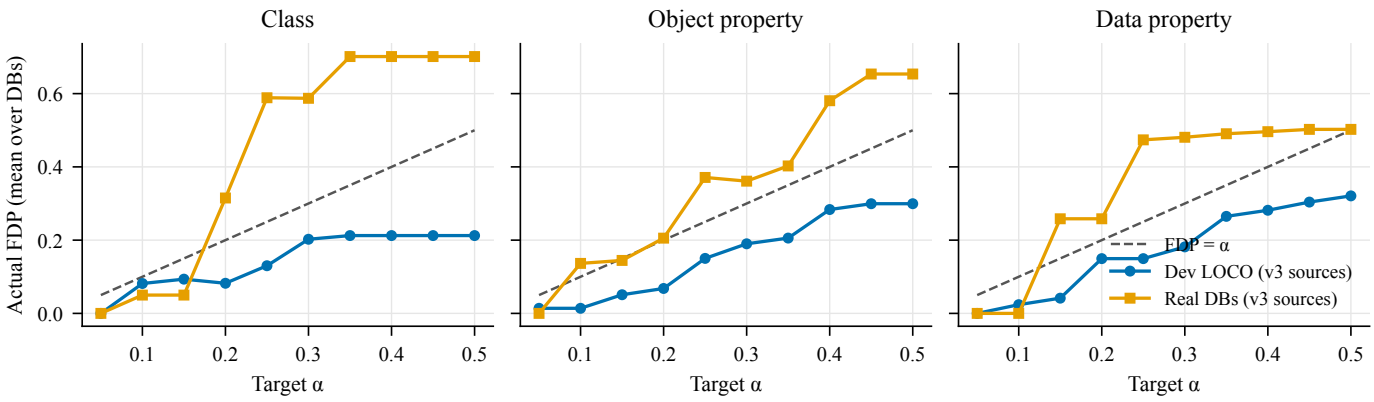


Fig. S12. CERT: realised false-discovery proportion (per-database mean) against the target α, original computation ("v3 sources"). Blue: leave-one-category-out calibration on the development set. Orange: the held-out real databases with the same thresholds. The dotted line is FDP $= \alpha$. The real-database pools contain the pre-fix whole-schema NPD outputs and the per-table outputs of the defective builder. The curves have not been redrawn after the post-hoc correction of the whole-schema sources; its $\alpha = .2$ values are given in Section VI-D and Table S12.

S3. Paired Complete-DDL Experiment: Details

Design: Three models (`qwen3.8-max`, `deepseek-v4.1-flash` and `kimi-k3`, BaiLian OpenAI-compatible route, temperature 0, `enable_thinking=false`, at most 65,536 output tokens, streamed with usage reporting) mapped each of the eight databases with data under two regimes, three times each. Per-table prompts hold the complete CREATE and constraint statements of the target table and of all its declared foreign-key neighbours, with no cap, or of the three tables with the most similar names when the target has none; whole-schema prompts hold the complete schema. No prompt contains instance rows, inclusion dependencies or gold information. On NPD, a third arm uses the defective builder's DDL for the target and related tables and is otherwise identical to the complete per-table arm, so the difference isolates the DDL bundle as a whole; it cannot separate the roles of column names, types and CREATE syntax. For the first two models, Burr's three databases took 198 prompts (3 whole-schema, 125 per-table and 70 defective-builder per-table) and 1,188 requests, and the five added databases took 53 prompts and 318 requests. Kimi-K3 ran the 181 complete prompts of both batches three times (543 requests) and did not run the defective-builder arm.

Execution and records: Each request was frozen before any call, with a hash of its body, model label, parameters and run index, and could be claimed only once; an uncertain outcome would have been recorded and never re-sent. The first two models ran in separate serial lanes, at most two requests in flight. All 1,506 of their requests returned HTTP 200 with a normal stop and a returned model label equal to the requested one; no request was retried. Two launch attempts of the second batch stopped before any request was claimed, because of malformed approval files on our side; they sent nothing and are kept in the records. For every request we keep the request bytes, the raw streamed response, the transport log and the usage report, and both batches were sealed by hash before normalisation. Fragments were normalised and merged once, in fixed table order. Truncated, empty or unparseable outputs would have stayed in the denominator as empty fragments; none occurred. For these two models reasoning-token counters were absent in all receipts, so we do not report their hidden reasoning as zero.

Kimi-K3 was added after the first two models had been scored. Its requests reused the frozen request bodies with the model label changed and went to the same route from a different client, which logged every attempt together with its raw stream. HTTP 429 responses would have been retried after a wait and transport failures at most twice, and truncated or empty outputs would have counted as they were; none of these cases occurred, and all 543 requests returned a normal stop with the requested label. Kimi-K3 reports a reasoning-token counter, which was zero in every receipt. This batch, too, was sealed by hash before normalisation, and normalisation, merging and scoring used the same frozen code.

Scoring: We scored the 222 whole-database mappings (102 of the first two models, 48 of Kimi-K3 and 72 of the key-column control below) with our scorer and, independently, with the mapping parser and metrics of Burr's official evaluator (without its database layer). The six F1 values and the element counts agree exactly for every mapping. The analysis plan was written after the calls returned and before any gold mapping was read, so we do not call it pre-registered.

Further results: Fig. S14 shows every run and element type, Fig. S15 the number of predicted elements relative to the gold, Fig. S16 the spread over the three runs, and Fig. S17 the token and latency cost. For Qwen and DeepSeek, over-generation underlies both large differences reported in the paper: whole-schema outputs on Mondial carry 2.1 times the gold attributes, most of them on key and join columns, and per-table outputs on BSBM 1.8 times the gold relations, mostly guesses for undeclared `publisher` columns. The merge statistics show that the per-table merge removed at most one attribute and one relation per mapping on these two databases, so the merge itself is not the driver. A post-hoc filter that removes data properties on URI-template or join columns from each mapping, applied to both regimes with no model call, brings the single call's Mondial attribute F1 to the per-table level (0.68 and 0.66 against 0.68 and 0.68 after the same filter) and changes the single call by at most 0.015 elsewhere; the remaining Mondial gap is in relations. At temperature 0, 42 of 48 cells give the same class F1 in all three runs. The largest changes are on NPD, where one Kimi-K3 single call left 15 of the 70 tables without classes or attributes and mapped 14 of them as many-to-many relations (class F1 0.63 against 0.87) and one DeepSeek single call returned 58 relations against about 100 in its other runs (relation F1 0.55 against 0.81).

Key-column control: The control adds one instruction to the eight whole-schema prompts, directly after the existing instruction to generate only as much content as the database needs: "Do not create a data property for a column that you already use in a class `id` pattern or in a `join`: key and foreign-key columns identify instances and link classes; they are not attributes." We wrote it after the Mondial analysis of the first two models and before any call of the control, and the rest of each prompt is unchanged byte for byte. Each of the three models ran each prompt three times (72 requests, all with a normal stop and none retried), and the outputs were sealed, normalised and scored like those above. Table S6 lists the three-way means, and Fig. S13 shows what the rule changes. The rule removes most key-column attributes for Qwen and DeepSeek on Mondial (115 to 4 and 109 to 25) and for Kimi-K3 on NPD (66 to 10), while Qwen keeps 69 of its 70 on NPD. It raises Qwen and DeepSeek on Mondial and lowers Kimi-K3 there; on the other databases it moves the single call by at most 0.025.

TABLE S6
KEY-COLUMN CONTROL: THREE-WAY MEAN F1 OF PLAIN WHOLE-SCHEMA GENERATION (W), WHOLE-SCHEMA GENERATION WITH THE KEY-COLUMN RULE (W+R) AND PER-TABLE GENERATION (P), MEAN OF THREE RUNS.

	Qwen3.8-Max			DeepSeek-V4.1-Flash			Kimi-K3		
Database	W	W+R	P	W	W+R	P	W	W+R	P
Bgee (6)	.59	.60	.59	.60	.60	.60	.60	.60	.59
ODH (8)	.53	.54	.50	.53	.54	.51	.54	.53	.53
ISWC (9)	.62	.63	.60	.63	.64	.49	.64	.64	.63
BSBM (10)	.91	.92	.84	.90	.90	.72	.91	.91	.87
GTFS (10)	.89	.89	.90	.88	.88	.87	.89	.86	.89
LUBM (14)	.15	.15	.14	.15	.15	.14	.15	.15	.15
Mondial (46)	.44	.51	.55	.46	.53	.60	.69	.66	.60
NPD (70)	.73	.73	.71	.70	.70	.69	.67	.67	.72
Mean (8 DBs)	.61	.62	.61	.61	.62	.58	.64	.63	.62

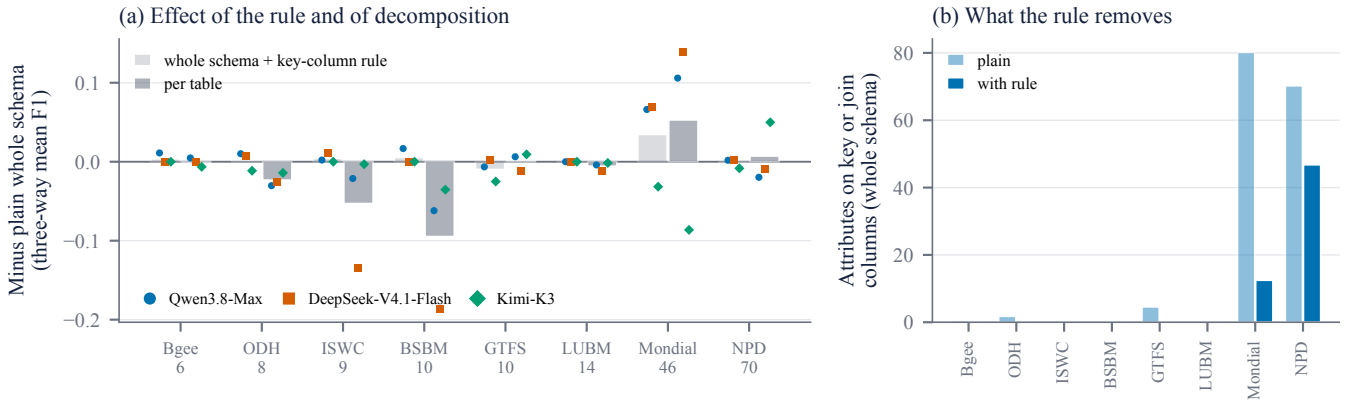


Fig. S13. Key-column control. (a) Change of the three-way mean F1 against plain whole-schema generation, with the key-column rule (light bars) and per table (dark bars); markers show the three models, bars their mean. (b) Data properties on URI-template or join columns in whole-schema outputs, mean of the three models, without and with the rule.

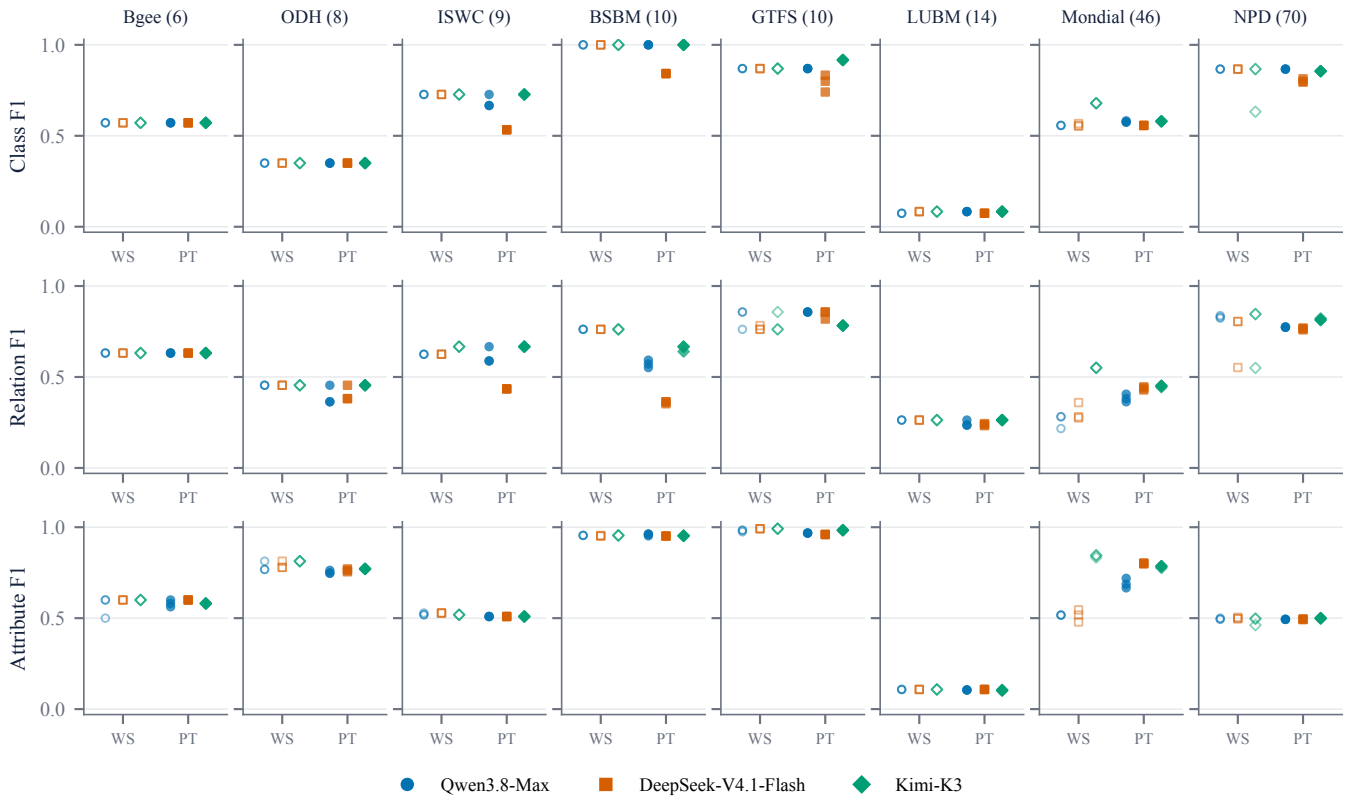


Fig. S14. Paired experiment, every run: mapping-based F1 per database and element type under whole-schema (WS, hollow) and per-table (PT, filled) generation, for the three models and three runs.

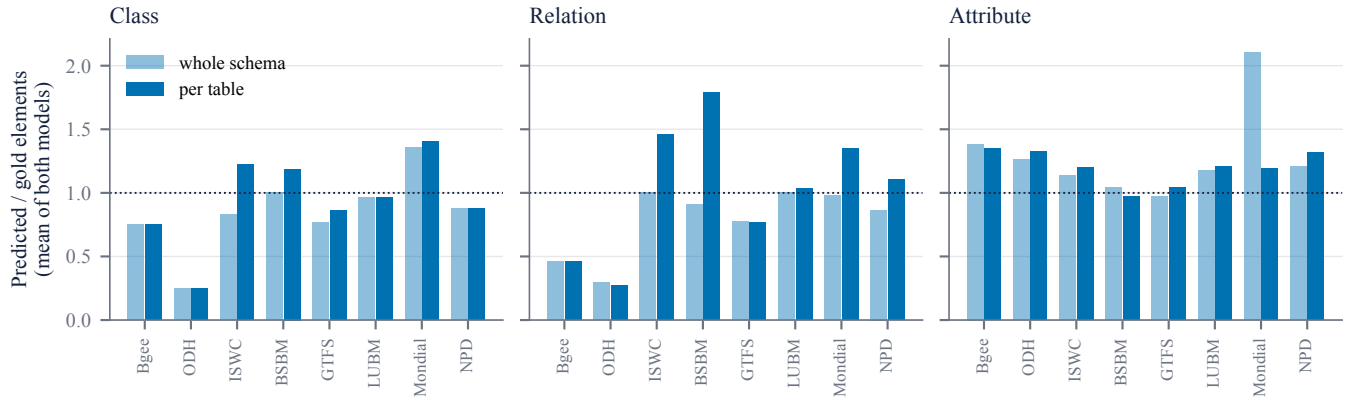


Fig. S15. Predicted elements relative to the gold, mean of Qwen and DeepSeek and three runs. Whole-schema outputs over-generate attributes on Mondial (2.1×); per-table outputs over-generate relations on BSBM and ISWC.

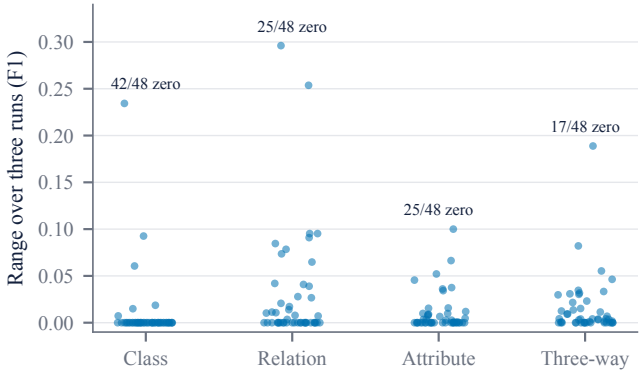


Fig. S16. Range of F1 over the three runs of each cell (3 models × 8 databases × 2 regimes), with the number of cells whose three runs agree exactly.

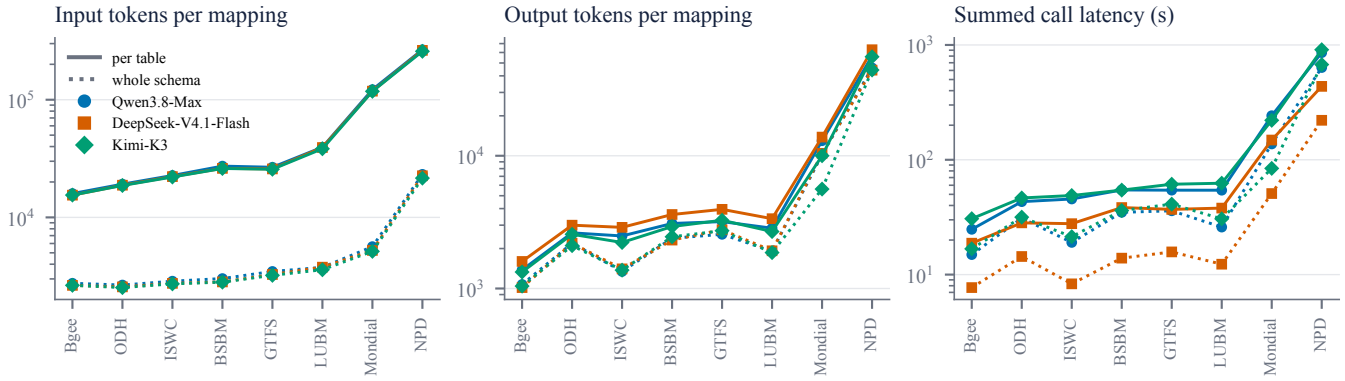


Fig. S17. Paired experiment, cost per whole-database mapping: input tokens, output tokens and summed call latency (log scale), per database, for per-table (solid) and whole-schema (dotted) generation. Summed latency is not wall-clock time.

Completion strategies: Three guards decide which tables to re-ask after a single call, without reading any gold mapping. G1 (planned) takes the tables that no class, data property or object property of the output mentions. G2 (planned, stricter) takes the tables without a class or a data property, even when a join mentions them. G3 (defined after seeing the scores of G1 and G2) takes the tables of G1 plus those that have a non-key column but no class or data property; key columns are primary keys, foreign keys and referenced columns, read from the DDL with the frozen parser. A re-asked table gets the archived per-table answer of the same run, since the per-table prompts do not depend on the single call. The answers are merged after the single-call output with the frozen per-table merge, so the single call keeps its elements in every conflict. G1 fired on 4 of the 72 single calls (19 tables), G2 on 34 (173 tables) and G3 on 14 (51 tables); Table S7 lists the eight-database results. DeepSeek's Mondial outputs had misspelt eight `geo_` table names and `riverthrough`; re-asking these tables added only wrong relations. Under G2 the per-table answers turn link tables into classes: DeepSeek loses 0.137 on ISWC (class precision .80 to .44) and 0.078 on BSBM, and Kimi-K3 loses 0.034 on Mondial. Merging without re-asking moves the eight-database means by less than 0.001, and taking the mean of the three per-table runs instead of the same run moves them by at most 0.001. Our scorer and the scoring code of Burr's official evaluator gave identical values for all 457 new mappings (292 for G1, G2 and merge-only controls, 114 for G3 and 51 with reasoning).

Reasoning mode: Qwen3.8-Max and Kimi-K3 ran the eight whole-schema prompts three times with `enable_thinking=true` (48 calls); apart from this switch the request bodies equal those of the paired experiment, which we checked by hash. All calls returned on the first attempt. Forty-five stopped normally, and the three Kimi-K3 calls on NPD stopped at the 65,536-token limit after 21.6k–26.4k reasoning tokens, which under the frozen rules counts as an empty mapping. As a diagnostic fixed before the calls, we re-sent these three with a limit of 131,072 tokens; they finished with 22.2k–32.4k reasoning and 44.5k–47.6k answer tokens, mapped every table and scored .731, .725 and .737. With reasoning, Qwen puts 15–20 of its 87 Mondial attributes on key or join columns (115 of 182 without), so relation F1 rises from .26 to .58 and attribute F1 from .52 to .79; on NPD every table gets a class but the outputs carry 0, 100 and 186 data properties (about 830 without), and attribute F1 falls from .50 to .12. Reasoning costs 2.5 (Qwen) and 1.9 (Kimi-K3) times the output tokens and 4.1 and 2.2 times the latency; even the six-table Bgee takes Qwen about 8.7k reasoning tokens and four minutes. The comparison uses the archived calls without reasoning, which were not re-sent at the same time (Table S8).

TABLE S7
COMPLETION STRATEGIES OVER THE EIGHT DATABASES: THREE-WAY MEAN F1, MEAN OF THREE RUNS; DIFFERENCES WITH 95% BOOTSTRAP INTERVALS OVER DATABASES; INPUT TOKENS RELATIVE TO THE SINGLE CALL. G3 WAS DEFINED AFTER SEEING THE SCORES OF G1 AND G2.

	Qwen3.8-Max	DeepSeek-V4.1-Flash	Kimi-K3
Single call / per table	.608 / .606	.606 / .576	.635 / .625
G1 / G2 / G3	.608 / .607 / .606	.605 / .576 / .599	.636 / .635 / .641
G1 − single call	+.000 [+.000, +.000]	−.001 [−.004, +.000]	+.000 [+.000, +.001]
G2 − single call	−.002 [−.006, +.001]	−.030 [−.067, −.003]	−.000 [−.013, +.013]
G3 − single call	−.002 [−.006, +.000]	−.007 [−.019, +.000]	+.006 [−.001, +.015]
G1 − per table	+.003 [−.032, +.030]	+.029 [−.033, +.094]	+.011 [−.012, +.037]
Input tokens, G1 / G2 / G3 / per table	1.00 / 1.46 / 1.02 / 11.4	1.34 / 1.88 / 1.39 / 11.6	1.02 / 3.01 / 1.63 / 11.8
Calls per database, G1 / G2 / G3 / per table	1.00 / 2.04 / 1.04 / 21.6	1.75 / 2.96 / 1.88 / 21.6	1.04 / 5.21 / 2.21 / 21.6

TABLE S8
REASONING OFF VERSUS ON: THREE-WAY MEAN F1 PER DATABASE, MEAN OF THREE RUNS; THE LAST ROW GIVES THE EIGHT-DATABASE DIFFERENCE WITH ITS 95% BOOTSTRAP INTERVAL. †ALL THREE CALLS REACHED THE OUTPUT LIMIT; WITH TWICE THE LIMIT THEY SCORE .731 ON AVERAGE, AND THE KIMI-K3 DIFFERENCE BECOMES −.003 [−.018, +.017].

Database	Qwen3.8-Max	Kimi-K3
Bgee (6)	.590 → .595	.601 → .597
ODH (8)	.529 → .529	.539 → .509
ISWC (9)	.625 → .588	.637 → .637
BSBM (10)	.906 → .898	.906 → .882
GTFS (10)	.892 → .901	.885 → .883
LUBM (14)	.148 → .152	.152 → .152
Mondial (46)	.445 → .686	.690 → .673
NPD (70)	.731 → .609	.674 → .000†
Eight databases	+.011 [−.046, +.087]	−.094 [−.262, −.005]

S4. LATER ROUND: NAMES, KEYS AND FURTHER MODELS

A later round of experiments was registered before any of its calls. Its renaming test has pre-registered hypotheses; the other analyses are exploratory. It reuses the prompts, the scorer and the $K = 3$ sampling of the main study. Primary comparisons use the eight databases with data; three further databases (MusicBrainz, UOBM and OMOP), whose converted gold mappings have known issues (Section S1), give secondary results. DeepSeek-V4.1-Flash is called through its official API with its default reasoning unless stated otherwise. Claude models do not accept a temperature setting and run with their defaults, and the GPT models run through a company gateway. Outputs truncated by an output limit or lost to a gateway time-out count as empty, as in the main study.

Identifier renaming: We built two variants of every database, one in which all table and column names are replaced by synonyms (paraphrased) and one with opaque identifiers, keeping data, structure and gold mappings aligned. Direct mapping, which ignores names, scores identically on all three variants, which checks the construction (Fig. S20). Hypothesis H-A1 held that, if the models do not rely on memorized names, paraphrasing lowers no LLM arm by more than 0.03 three-way mean F1. On the eight databases, the drops (original minus paraphrased) are -0.016, $+0.016$, -0.004 and -0.012 for whole-schema and per-table generation with DeepSeek and Kimi-K3 (one-sided Holm $p \geq .64$), so H-A1 holds; DET-V2 changes by -0.001. Hypothesis H-A2 held that opaque names hurt the LLM arms more than the rules. Opaque names lower the LLM arms by 0.017–0.059 and DET-V2, whose type rules read names, by 0.026; relative to DET-V2, only per-table generation with Kimi-K3 drops significantly more ($+0.015$, 95% interval $[0.002, 0.038]$, Holm $p = .021$). Names thus help both routes a little, and we find no sign that the models rely on memorized benchmark names.

Schemas without declared keys: Removing all primary, foreign and unique key declarations while keeping the data shows how much each route relies on them (Fig. S21). On the eight databases, the relation F1 of DET-V2 falls from 0.56 to 0.35 and that of direct mapping from 0.43 to 0, while the four LLM arms fall only from 0.56–0.61 to 0.53–0.60. The LLM arms then lead DET-V2 by 0.18–0.25 relation F1 and by 0.08–0.13 three-way mean F1 (two-sided Holm $p \leq .005$, also on all eleven databases). The rules recover keys from data, but on real databases not well enough to replace declared ones.

Sample rows: Giving the single call 0, 5 or 20 sample rows per table does not change average quality (all four comparisons Holm $p \geq .82$ on eight databases); with 20 rows the NPD prompt reaches about 339,000 tokens, and neither model fails.

Output completeness: Across all 9,673 whole-schema outputs of the project, 148 cover fewer than 90% of the tables. The share rises from 0.2% for schemas of 1–5 tables to 27% for 51–100 tables; in a logistic regression on log table count and log prompt length, table count carries the effect (coefficient 1.12, SE 0.13) and prompt length adds none. Incompleteness is model-specific (Fig. S19): on NPD, Claude Sonnet, Haiku and Fable and two of the GPT models never finish, whereas Claude Opus, Kimi-K3, Qwen3.8-Max and GPT-6-luna always do. For DeepSeek, the default reasoning consumes the output budget. With thinking explicitly disabled, its eight-database scores change by $+0.017$ (whole schema) and -0.024 (per table), neither significantly; its NPD single calls finish in 3 of 3 samples (1 of 3 with default reasoning) and its failed per-table calls drop from 31 of 741 to none.

Further models: Fig. S18 adds four Claude and three GPT models to the grid. On the development set the Claude models score 0.81–0.84, above the other models (0.77–0.80) and below the rules (0.87). On the seven held-out databases other than NPD, all thirteen whole-schema configurations lie within 0.595–0.627 (DET-V2: 0.594); on all eight they span

0.52–0.64, almost entirely because of NPD. The best configuration, Claude Opus 5.5, scores 0.640 on eight databases ($+0.029$ over DET-V2, Holm $p = .37$) and 0.560 on eleven ($+0.066$, Holm $p = .007$). Per-table generation matters most for models whose single calls are truncated or time out: on NPD, Claude Sonnet, Claude Haiku and GPT-6.1-sol score 0.00, 0.17 and 0.00 in a single call and 0.73, 0.69 and 0.72 per table. For Claude Opus, which finishes, per-table generation scores lower (eleven databases -0.026, Holm $p = .019$).

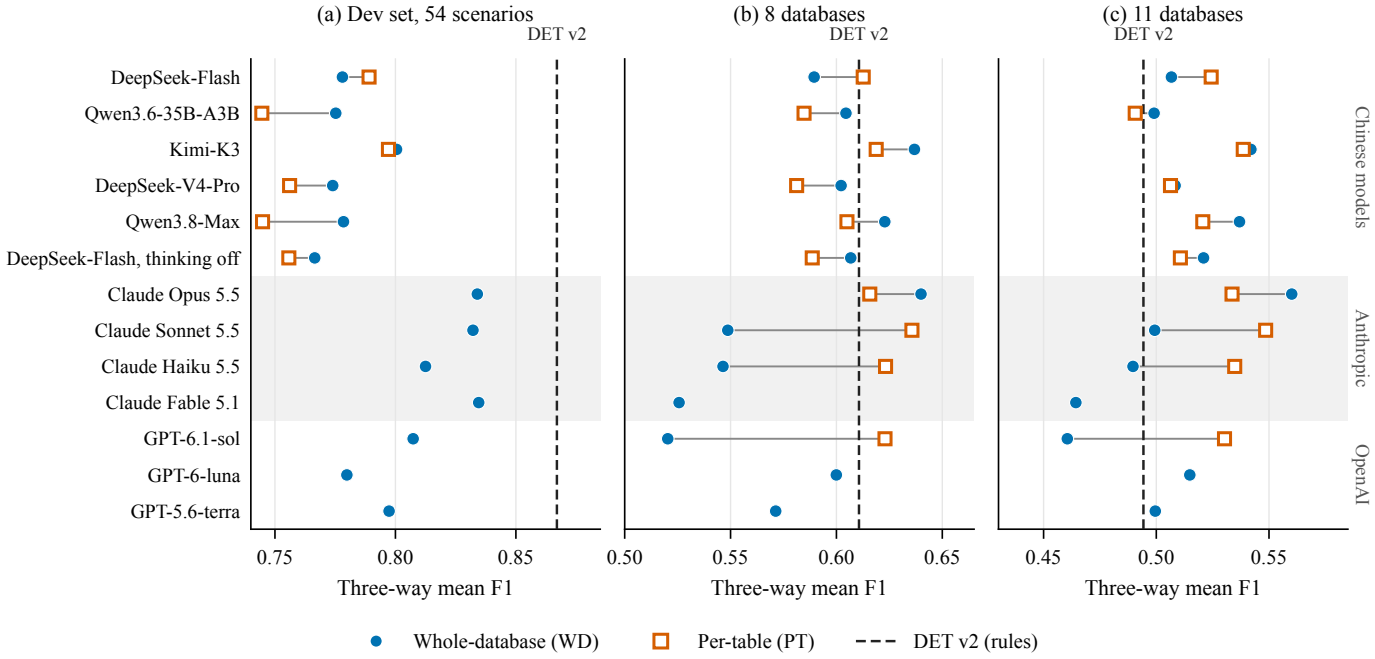


Fig. S18. Model grid, three-way mean F1: (a) development set, (b) eight held-out databases, (c) eleven databases including MusicBrainz, UOBM and OMOP, whose gold mappings have known issues. Circles: whole-schema generation; squares: per-table generation; dashed line: DET-V2. DeepSeek-Flash uses its API's default reasoning unless marked as thinking off.

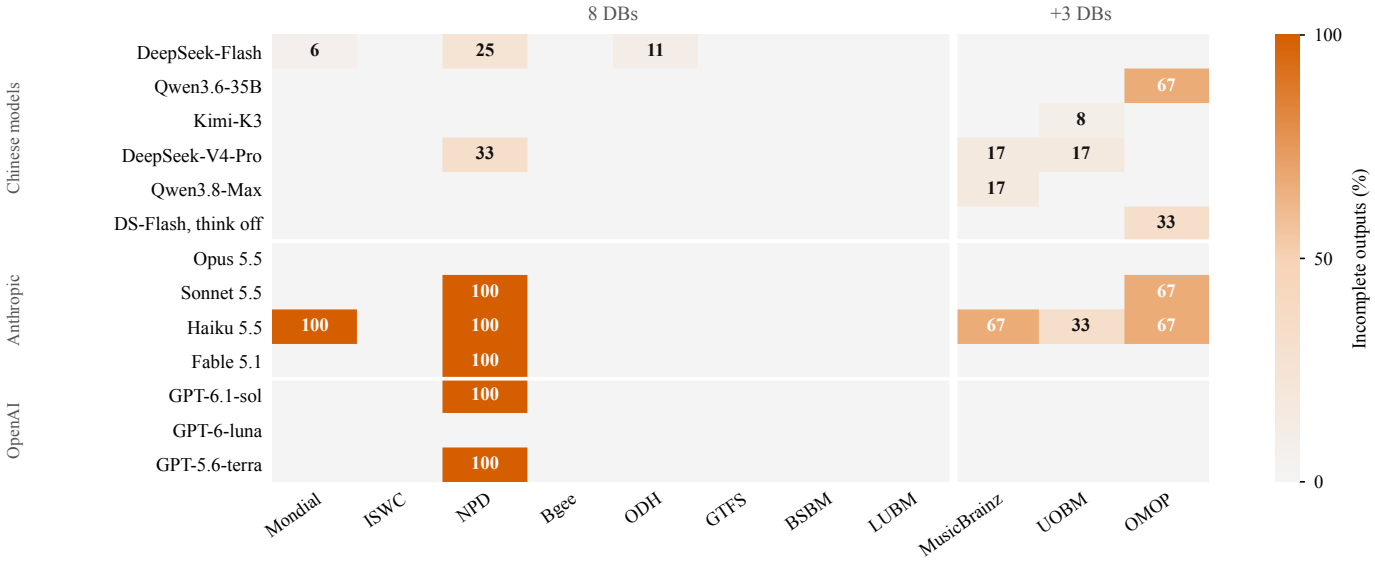


Fig. S19. Share of incomplete whole-schema outputs (coverage below 90% of the tables) by model and database, over all prompts and samples. Most cells are empty; incompleteness concentrates on NPD and on a few models.

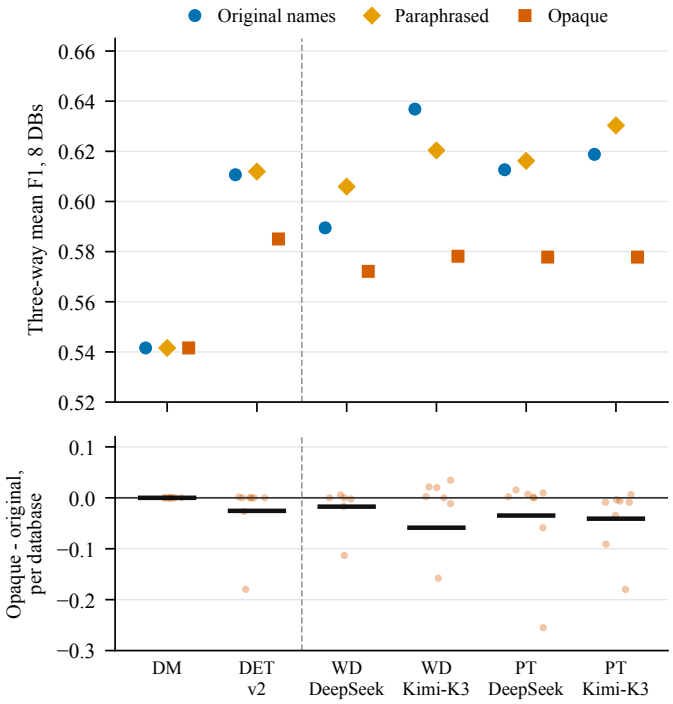


Fig. S20. Identifier renaming on the eight databases. Top: three-way mean F1 with original, paraphrased and opaque names. Bottom: opaque minus original per database (dots) and on average (bars). Direct mapping ignores names and is unchanged.

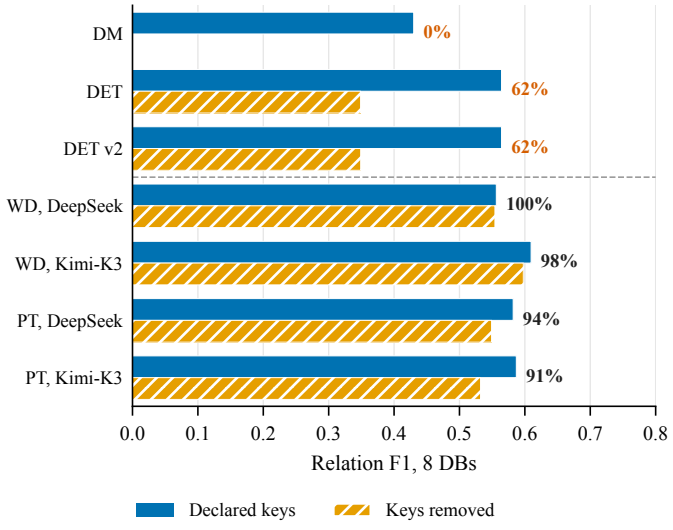


Fig. S21. Relation F1 on the eight databases with declared keys (solid) and with all keys removed (hatched); percentages give the ratio of the two eight-database means.

TABLE S9
EXECUTABLE CHECK OF THE CONVERTED GOLD MAPPINGS: RETAINED ENTRIES WHOSE SOURCE AND CONVERTED FORMS RETURN IDENTICAL VALUES ON THE DATA, VALUES THAT DIFFER ONLY IN PUNCTUATION, AND EMPTY RESULTS ON BOTH SIDES.

Database	Entries	Identical	Punctuation	Both empty
Bgee	34	23	11	0
ODH	91	91	0	0
GTFS	86	79	0	7
BSBM	72	72	0	0
LUBM	49	49	0	0
All	332	314	11	7

TABLE S10
MODEL-BASED AUDIT OF THE CONVERTED GOLD MAPPINGS: ENTRIES PER LABEL AFTER ADJUDICATION, AND AGREEMENT OF THE TWO INDEPENDENT PASSES.

Database	Entries	Correct	Approx.	Wrong	κ
Bgee	34	31	3	0	1.00
ODH	50	47	3	0	1.00
GTFS	50	45	5	0	0.88
BSBM	50	46	4	0	0.88
LUBM	49	31	4	14	0.46
All	233	200	19	14	0.71

S5. CHECKS OF THE CONVERTED GOLD MAPPINGS

Executable check: We ran every retained entry of the five converted gold mappings and its source mapping on the same DuckDB data and compared the returned values: key tuples for a class, (subject key, value) pairs for a data property and (subject key, object key) pairs for an object property. The source side executes the original SQL and reads the template variables; the converted side uses the extent metric's D2RQ assembly. Only values are compared, so a variable resolved to the wrong column would show. Of the 332 entries, 314 return identical values (Table S9). The 11 Bgee entries that differ only in punctuation encode identifiers with `replace(id, ':', '_')` in the source, which the converted gold marks as `|urlify` and reads from the original column; one of them is a concatenation that lost its separator. The 7 GTFS entries that return nothing on either side map columns that are empty in the generated data. The check proves equivalence on this data only, and both sides share one mapping parser, so a misread source or an approximation hidden by the data would pass it.

Model-based audit: A language model audited the converted gold mappings of the five added databases. The sample is the review queue drawn with a fixed seed (20261024) from the converted gold without reading any system output: all 34 retained entries of Bgee, all 49 of LUBM and 50 each of ODH, GTFS and BSBM, that is 233 of the 332 retained entries. Two independent passes of Claude Opus 5.5, run as separate agents, labelled every entry as *correct*, *approximate* (a describable loss that keeps the target concept), *wrong* (a changed meaning) or *unclear*, following a written rubric that lists the converter's lossless and approximate rules. Each pass could open the source mapping, the schema and the converted gold, but neither the other pass's labels nor the converter's own flags; the second pass saw the entries in shuffled order. The passes agreed on 217 of the 233 entries (Cohen's $\kappa = 0.71$; 0.93 without LUBM), and a third Opus 5.5 session resolved the 16 disagreements by reading the sources (Table S10).

All 14 wrong entries are in LUBM and share one cause. LUBM's R2RML mapping gives names, e-mail addresses, degrees, teaching, research interests and advisors through subject maps that have no class and share their IRI template with Lecturer and the three professor classes. Our converter attached these properties to Lecturer alone, which shrinks their subjects from all faculty to lecturers; its own rules call for marking such entries inexpressible or attaching them to every matching class. Each of these entries keeps the columns, joins and conditions of the original SQL, and these are exactly what Burr's mapping-based metric compares; the class that a property belongs to does not enter the comparison. Re-attaching all 14 entries to another class left every mapping-based and name-based F1 unchanged for all 150 LUBM outputs we re-scored: the 123 in the project's output folder from the main study and the later round, and the 27 of the paired experiment and its key-column control. This is a concrete case for scoring each property together with its class (Section VII). We fixed the entries anyway, because the extent metric reads the class. When a source map declares no class and several class maps share its IRI template, the converted entry keeps Burr's single class field and also lists all those classes, and the extent metric takes the union; 16 LUBM and 6 ODH entries carry such a list, and after the fix all 49 LUBM entries return the same values as the source (Table S9). The added lists are invisible to Burr's parser: re-scoring the 58 LUBM and ODH outputs of the eight-database comparison changed no mapping-based or name-based F1. Under the extent metric, LUBM attribute F1 drops by 0.001–0.002 for every output, because one property whose extent was empty now counts, and no eight-database mean moves by more than 0.001; the paper reports the values before the fix.

The 19 approximations are 12 IRI objects without a matching class that became literal attributes (GTFS code lists, BSBM publishers and countries, Bgee cross-references, ODH geometry nodes), four LUBM relations whose object IRI takes its department from the subject, so that it matches the target class only when the two departments coincide, one Bgee concatenation that lost its separator, one ODH join that lost a zero-padding function and one GTFS self-join that the format cannot alias. The BSBM publishers bear on the paired experiment: the gold maps these undeclared `publisher` columns as attributes, so per-table outputs that model them as relations lose relation precision (Section VI-B3). The converter's own approximation flags covered 26 of the 33 entries that the audit judged approximate or wrong. Both passes use the same model, so their agreement measures the consistency of that model.

S6. BURR BASELINES: PUBLISHED AND CORRECTED

Table S11 sets the published values that we quote next to the corrected values that the Burr authors report in the Updated Evaluation Results of their repository README, pinned at

TABLE S11
MAPPING-BASED F1 (CLASS/RELATION/ATTRIBUTE) OF BURR'S RULE-BASED BASELINES: PUBLISHED VALUES AND THE AUTHORS' CORRECTED VALUES [4], WHICH TABLE I USES. NEITHER ROW IS A RUN OF OURS.

Database	System	Published†	Corrected (README)
Mondial	Ontop	.53/.35/.51	.56/.28/.51
Mondial	RDB2Onto	.49/.24/.48	.54/.31/.67
NPD	Ontop	.68/.84/.48	.85/.85/.48
NPD	RDB2Onto	.60/.81/.48	.70/.81/.49

commit `530a5bc` [4]. The README attributes the update to the correction of a small number of missing constraints in Mondial and NPD and states that the evaluation setup and the relative system rankings are unchanged. It also reports that one empty table was removed from the Enterprise data, leaving seven tables, and it lists corrected lexical scores, which this paper does not use. The corrected values are those reported by the authors; we did not re-run these systems. Our database files and Burr's evaluator are byte-identical to this release: the git blob hashes of all 353 tracked files match commit `530a5bc`, and no commit after the repository's first one changes a data file, so our own systems ran on the corrected constraints. Table I therefore uses the corrected Ontop and RDB2Onto values for Mondial and NPD and recomputes their averages; no corrected values exist for Gemini, whose published values stay. The critical-difference diagram (Fig. S8) still uses the published values.

S7. RISK-CONTROL SOURCE CORRECTION

Table S12 compares the original computation with the post-hoc correction of Section VI-D. The corrected run reuses cached outputs and calls no model. Its per-table and HYBRID2 NPD sources from the defective builder have not been replaced with verified complete-DDL outputs. Closing this gap requires nine corrected per-table NPD outputs for the 46-source pool, twelve for the 49-source variant and six for the 27-source variant, and the HYBRID2 sources share some of these per-table dependencies; the source sets overlap, so these counts must not be added. Neither the 12 outputs of the complete-DDL control nor the outputs of the target-only probe can fill the gap: the control is a separate two-model cohort built with the defective per-table builder, and the probe covers only 12 tables and adds only the target's CREATE statement. The $\alpha = .2$ values in Table S12 are the post-correction results reported here. Other WD-only base and $\alpha = .1/.2$ points have been computed offline but are not added to this table. They do not supply the missing complete-PT/HYBRID2 source correction, full curves or leave-one-category-out analysis; the full figure retained here is the original computation. The guarantee of Section IV-D concerns the expected per-database FDP, so the database mean is the relevant comparison with α; the pooled values weight databases by their number of accepted elements.

TABLE S12
CERT AT $\alpha = .2$ ON BURR'S FOUR HELD-OUT DATABASES, 46-SOURCE POOL. ORIGINAL: FIRST COMPUTATION (FIG. S12). CORRECTED: POST HOC, WITH ONLY THE WHOLE-SCHEMA SOURCES REPLACED BY THEIR CORRECTED-PROMPT OUTPUTS. DB MEAN: MEAN OF PER-DATABASE FDP; POOLED: WRONG OVER ACCEPTED ELEMENTS ACROSS DATABASES. ALL DATABASE MEANS EXCEED α.

	Original	Corrected (whole-schema sources)			
Element	DB mean	DB mean	Pooled	Accepted	Wrong
Class	.315	.317	.221	122	27
Relation	.205	.232	.215	158	34
Attribute	.258	.222	.265	113	30

S8. TARGET-ONLY CREATE PROBE

Before the paired experiment, a small post-hoc probe inserted only the target table's CREATE statement into affected r5 per-table prompts for NPD and checked whether the target's class map changed. Table S13 gives its design, configuration, completion status and results. The 12 target tables were selected from the 66 NPD tables whose per-table prompt lacked a CREATE statement. The legacy prompt is the r5 per-table prompt in its plain variant, without class names from earlier tables, with five sample rows per table and the historical related-table context. The two inserted blocks differ only in content: a minimal CREATE statement that declares the target's columns and types, or a fixed SQL comment of equal length in characters. Equal characters do not mean equal tokens. Because the related-table context stays incomplete in every condition, the probe tests the inserted block as a whole and cannot separate the roles of names, types and CREATE syntax. Outputs were averaged over the three repeats of each table before the contrasts were formed. The 12 tables come from one database and are not independent replications, so the table-level bootstrap interval is descriptive. Unmatched means unmatched under Burr's matching; no person checked these attributes. Each of the 108 slot-1 receipts matches the hash of its frozen request; every call returned HTTP 200 with a normal stop and a completed stream, and none showed reasoning text. The reasoning-token field is null in all 108 receipts, so we do not report it as zero, and the returned label is the provider's and does not verify the served weights. For the second slot, a reported count of zero reasoning tokens does not prove that the upstream non-thinking contract was executed, and the route audit does not identify the weights served. The five requests were not retried, the stop was not cleared, and the 103 unsent positions count neither as errors nor as successes. The probe is not a corrected per-table arm and enters no table or candidate pool of the paper. Its token usage is listed in Table S17.

TABLE S13
TARGET-ONLY CREATE PROBE ON NPD: DESIGN, CONFIGURATION, COMPLETION AND RESULTS.

Item	Value
Design	12 of the 66 affected NPD target tables × 3 conditions (legacy, target-CREATE, sham) × 3 repeats × 2 model slots = 216 planned calls; 113 issued, 103 not issued
Context	r5 per-table prompt, plain variant: no earlier class names, historical related-table context, five sample rows per table
Inserted block	Target-CREATE: minimal CREATE with the target's columns and types. Sham: fixed SQL comment of equal character (not token) length
Slot 1	`qwen3.8-max`, requested and returned; BaiLian workspace, OpenAI-compatible route; temperature 0.2; at most 8,192 output tokens; `enable_thinking=false`; streamed with usage reporting. 108 issued, each sent once, all HTTP 200 with normal stop; reasoning-token field null in all 108
Slot 2	GPT-family route: 5 issued (3 mechanically successful, 2 returned a thinking-mismatch receipt), none eligible; 103 never sent
Primary endpoint	Target-owned class map, slot 1: 1 correct and 0 unmatched in each of the 36 outputs per condition
Contrasts	Target-CREATE − legacy = 0; target-CREATE − sham = 0; table-level bootstrap interval $[0, 0]$
Diagnostic	Direct-column attributes, predicted/matched/unmatched: legacy 451/226/225; target-CREATE 468/229/239; sham 446/228/218

TABLE S14
CONFIGURATIONS. COST IS THE LLM API CHARGE PER NPD SAMPLE RECORDED IN THE MAIN STUDY AT THE PRICES OF THE TIME; RULE-BASED ARMS INCUR NO MODEL CHARGE, BUT PROFILING HAS A COMPUTATIONAL COST (SECTION VI-F).

System	Input used	Model calls	Cost (NPD)
DM	declared keys	0	0
DET, DET-V2	profiling evidence	0	0
WHOLE-DB	schema + 5 rows/table	1	$0.04
PER-TABLE	per-table context	1 per table	$0.85–0.91
HYBRID2	DET ∪ PER-TABLE + gates	1 per table	as PER-TABLE
CERT	all of the above	0 extra	0

S9. MODEL-CALL CONFIGURATIONS BY COHORT

Table S14 lists the systems of the main study, each a path through the components of Fig. 2.

Table S15 separates the call settings of the main study from those of the later runs. Labels are those requested from and returned by the providers. None of them verifies the weights that were served, and a requested setting is not proof that a route applied it.

TABLE S15
MODEL-CALL CONFIGURATIONS BY COHORT, WITH WHAT THE AVAILABLE RECORDS DO NOT ESTABLISH.

Cohort	Labels, settings and limits
Main study (main tables and figures)	The nine models of Section III, DeepSeek V4.1 Flash as reference; temperature 0; K samples told apart by a sample id; DeepSeek per-table calls capped at 8,192 output tokens (Section VI-B1). Not established: served weights.
CREATE probe, slot 1 (Section S8)	`qwen3.8-max`, requested and returned; BaiLian workspace, OpenAI-compatible route; temperature 0.2; at most 8,192 output tokens; `enable_thinking=false`; streamed with usage reporting; r5 per-table context, plain variant. Not established: hidden reasoning (counters null), served weights.
CREATE probe, slot 2	GPT-family route; frozen non-thinking, temperature and token-limit contract requested. Not established for any of the five issued responses, so all five are excluded.
Vendor track (Section S10)	`oa:gpt-6-luna` and `oa:gpt-5.6-terra`; returned labels match in 741 of 741 table responses each; temperature 0; at most 8,192 output tokens; no reasoning setting sent; r5 per-table context, ctx variant; $K = 3$. Not established: parameter pass-through, non-thinking execution (no reasoning counters), served weights.
Vendor track, third configuration	Fable (label as recorded): 28 of 33 database samples, not averaged.
Later round (Section S4)	`deepseek-v4.1-flash` through its official API with default reasoning, and with thinking explicitly disabled as a separate configuration; BaiLian models as in the main study; Claude Opus, Sonnet and Haiku 5.5 and Fable 5.1 at their default temperature, which cannot be set; GPT-6.1-sol, GPT-6-luna and GPT-5.6-terra through a company gateway; truncated or timed-out outputs count as empty. Not established: served weights.
Paired experiment and key-column control (Section VI-B3)	`qwen3.8-max`, `deepseek-v4.1-flash` and `kimi-k3`, BaiLian OpenAI-compatible route; requested and returned labels match in every call; temperature 0; at most 65,536 output tokens; `enable_thinking=false`; streamed with usage. Prompt: complete CREATE and constraint statements of the target table and of all its declared foreign-key neighbours (three name-similar tables when it has none), no rows; whole-schema prompts hold the complete schema. Every request sent once, no retries. Not established: hidden reasoning of the first two models (counters absent; Kimi-K3 reports zero), served weights.

S10. EXPLORATORY VENDOR PER-TABLE TRACK

This section reports two completed configurations of the later extension (Section S1). They are outside the main study's nine-model ladder and enter none of the main tables, figures, tests or candidate pools, and we report no p-values for them. Both were run with the r5 per-table regime in its ctx variant (Section IV-B), with $K = 3$, on the eleven databases of the extension: the eight held-out databases of Table S4, whose identifiers match, plus MusicBrainz, UOBM and OMOP; Enterprise is excluded. Each configuration completed 33 of 33 database samples, with all 741 expected table responses and no failed-table marker in the saved predictions. Table S16 lists the saved scores: mapping F1 of the r5 scorer, averaged equally over class, relation and attribute within each database sample and then over the 24 (eight databases) or 33 (eleven databases) database samples. We checked that the 66 prediction files are bound by hash to these saved scores. We did not rescore them, and they are not established as part of the replay of

TABLE S16
EXPLORATORY VENDOR TRACK, DESCRIPTIVE AND OUTSIDE THE MAIN STUDY: SAVED THREE-WAY MEAN MAPPING F1, $K = 3$. WD: HISTORICAL CACHED WHOLE-SCHEMA OUTPUTS. PT: R5 PER-TABLE, CTX VARIANT, WITH THE NPD INPUT DEFECT. VALUES ARE ROUNDED ONCE TO THREE DECIMALS; DIFFERENCES ARE COMPUTED BEFORE ROUNDING. —: INCOMPLETE, NOT AVERAGED.

	`oa:gpt-6-luna`	`oa:gpt-5.6-terra`	Fable
Database samples	33/33	33/33	28/33
Table responses	741	741	—
8 DBs, WD	.600	.571	—
8 DBs, PT	.620	.606	—
11 DBs, WD	.515	.500	—
11 DBs, PT	.536	.528	—
NPD, WD	.520	.291	—
NPD, PT	.717	.711	—
NPD, PT − WD	+.197	+.419	—

Section VII. The whole-schema values are historical cached outputs of the same labels and were not called at the same time as the per-table ones.

Five boundaries govern how these numbers can be read. First, the eight database identifiers match Table S4, but the prompts, gold files and versions have not been shown to be byte-identical to those behind the original predictions. Second, the per-table prompts carry the NPD defect of Section VI-B1: 66 of the 70 tables have no CREATE statement. Third, the eleven-database means include the OMOP and MusicBrainz gold issues of Section S1. They complement and never replace an eight-database mean, and the three databases they add are not databases of this paper's cohort. Fourth, the call settings and their limits are those of Table S15: no reasoning setting was sent, no reasoning counters were recorded, and the batch retried after transport failures (Table S17). Fifth, development-set values of this track use a different aggregation from the original ladder and are not merged into Fig. 3. Within these boundaries, the stored NPD differences between per-table and whole-schema means are +.197 and +.419, outside the range from −.13 to +.08 that holds for the main study everywhere except DeepSeek on NPD (Section VI-B2). The two regimes differ here in source and model versions and in retry behaviour as well, so these differences do not isolate a decomposition effect. The third configuration (Fable) completed 28 of 33 database samples; a resume for the remaining five (93 table positions) was blocked, and we leave its averages blank without imputation.

S11. RESOURCE ACCOUNTING

Table S17 lists the resources of each cohort in its own units. Database samples, table-level responses, HTTP attempts and model calls are different units and must not be added across rows. Token counts are the usage reported by each route, whose accounting of reasoning, caching and billing differs, so they are not an equal-compute measure. We do not know the monetary cost of the later runs, and we give no total for the whole project: historical and later records would first have to be deduplicated where they share cached outputs, and the boundary of the experiment inventory would have to be fixed. Two scope notes apply to the paper. Rule-based arms incur no LLM charge, but profiling costs computation (Section VI-F). CERT makes no extra calls once its source pool exists, but the pool itself required the calls of the source arms.

TABLE S17
RESOURCE ACCOUNTING BY COHORT. IN/OUT: REPORTED PROMPT AND COMPLETION TOKENS. ROWS USE DIFFERENT UNITS AND ARE NOT ADDITIVE.

Item	Record
Main study: unreconciled run records	
DeepSeek	about $67
Seven further models	7,317 calls; about 44M input tokens
Per NPD sample	WHOLE-DB about $0.04; PER-TABLE $0.85–0.91, at the prices of the time
Vendor track (Section S10); monetary cost unknown	
Database samples	94 of 99 planned (33, 33 and 28)
Recorded responses	2,131, one of them failed; in 15,554,756, out 1,451,385; usage of disconnected requests may be missing
`gpt-6-luna`	741 successful responses from 742 HTTP attempts (741 with status 200, one network exception); in 4,891,056, out 356,570
`gpt-5.6-terra`	741 successful responses from 746 HTTP attempts (741 with status 200, three network exceptions, two with status 500); in 4,944,122, out 586,310
Delivery	retried after transport failures, so not exactly-once
CREATE probe (Section S8); monetary cost unknown	
Calls	216 planned; 113 issued, each sent once; 103 not issued and not counted
Slot 1, 108 calls	in 1,596,108, out 82,673, total 1,678,781
Slot 2, 5 calls	in 87,735, out 4,381, total 92,116; consumed although excluded
Combined	in 1,683,843, out 87,054, total 1,770,897
Paired experiment (Section VI-B3); monetary cost unknown	
Requests	1,506 planned, 1,506 sent once each; all HTTP 200, normal stop, returned label equal to the requested one; no retries
Kimi-K3, 543	in 1,698,879, out 426,377; all sent once, normal stop
Key-column control, 72	`qwen3.8-max` in 143,283, out 190,788; `deepseek-v4.1-flash` in 138,024, out 187,617; `kimi-k3` in 133,770, out 187,484
Burr's three databases, 1,188	`qwen3.8-max` in 1,987,179, out 454,219; `deepseek-v4.1-flash` in 1,961,511, out 484,776
Five added databases, 318	`qwen3.8-max` in 432,666, out 69,773; `deepseek-v4.1-flash` in 421,401, out 77,188
Reasoning tokens	field absent in all 1,506 receipts of the first two models and in their control receipts; zero in all 567 Kimi-K3 receipts
Later round (Section S4); ledger estimates	
DeepSeek	5,591 calls (default reasoning and thinking disabled), about $27
Kimi-K3	2,385 calls, at most CNY 590 at list price
Claude	3,023 calls; about $91 for Opus, Sonnet and Haiku; Fable price not recorded
GPT	1,324 calls through a company gateway, not billed

S12. AUDIT OF REPORTED NUMBERS

Eight-database table (Table S4): The values of Table S4 are computed in three steps: the mean over samples within each database, the equally weighted mean over databases, and a single rounding to two decimals. Seven cells of the previous version did not follow this rule and were one unit too high. Table S18 lists them with their unrounded values. The five corrected class cells on the added databases share one

TABLE S18
DISPLAY CORRECTIONS IN TABLE S4. OLD: PREVIOUS VERSION; UNROUNDED: THE THREE-STEP AGGREGATE DESCRIBED IN THE TEXT; NEW: THIS VERSION.

System	Column	Element	Old	Unrounded	New
DET	Added 5	class	.58	.5749	.57
DET-V2	8 DBs	class	.64	.6349	.63
DET-V2	Added 5	class	.58	.5749	.57
HYBRID2	8 DBs	attr.	.63	.6245	.62
WHOLE-DB DeepSeek	Added 5	class	.58	.5749	.57
WHOLE-DB 35B-A3B	Added 5	class	.58	.5749	.57
WHOLE-DB Kimi-K3	Added 5	class	.58	.5749	.57

unrounded value, in line with the observation of Section III that class F1 there is largely fixed by the gold structure. Four ISWC relation cells of the main table (Table I: DET, DET-V2, HYBRID2 and whole-schema DeepSeek) have an unrounded value of exactly .625. They remain at .62, rounded half to even as before; we did not change them in isolation, because a different tie rule would have to be applied to the whole paper. No underlying value was changed.

Complete-DDL control (Table S5): An independent recomputation proposed .40, .71 and .74 for cells of Table S5. These values come from outputs of a different production queue and do not apply to the frozen 12-output control of this table. We retain the control values .59, .70, .68 and .72 (unrounded .594, .702, .676 and .722) and its class counts: 70/70/20 and 70/70/70 for whole-schema generation, 72/68/68 and 70/70/70 for per-table generation, for DeepSeek and Kimi-K3 respectively. The control is exploratory, and we do not reuse it for the corrected CERT pool or for the probe of Section S8.

Earlier recomputation and its scope: Before the replay of Section VII, an independent recomputation compared 603 numerical cells of an earlier manuscript snapshot with values recomputed from the frozen outputs. Of these, 522 matched, 18 were flagged, and 63 cells that report other authors' results were not recomputed; the replay reproduced this outcome. Five flags concern Table S5 and are superseded as explained above, and the seven corrections of Table S18 were adjudicated against unrounded values. The adjudication of the remaining flagged cells is not covered by the evidence available for this version. Neither the recomputation nor the replay checks how the outputs were generated, which input version they used, or whether the gold mappings are semantically correct.

Frozen-input replay: The earlier internal audit cohort is frozen in a manifest of 11,886 unique input files (about 1.0 GB). It does not cover every computation subsequently added to this paper. It holds the raw predictions of 4,735 scoring tasks and 9,673 coverage tasks, together with the scorer, gold mappings, schemas, prompt bundles and instance databases they need; tasks share files, so task counts are not file counts. The computations were executed three times: by an earlier independent recomputation that read the original directories, from an independent local copy of the manifest, and in a container without network access and with a read-only file system. The eight core result files, including the per-task scores, the coverage census, the main table and the significance families, are byte-identical across the three executions. The extent results for the three instance databases agree within 10^{-12} in every field except timings. All three executions record the same 148 incomplete outputs, 62 call or parse failures and 8 scoring errors; we keep all of them. All 26 automated checks of the package pass, including checks for input tampering, extra gold files, writes to the sources, network access, path escapes and leaked secrets. The later post-hoc CERT correction, the CREATE probe and the new vendor predictions are outside this replay guarantee. The replay called no model and ran no generation arm. It reproduces completed computations from frozen inputs; it is not a new experiment and not a validation of the gold mappings, and it is distinct from the 640-unit check against the official evaluator (Section V). Section S12 records what it does not cover, including 63 externally reported cells that were not recomputed. The 66 prediction files of the later vendor track (Section S10) were checked only for agreement between their hashes and the saved scores; they were not rescored, and they are not established as part of the replayed task set.

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